VOUCH

Developer Master Documentation

Fair & Explainable Creditworthiness Assessment (F-04)

Architecture • stack • graph model • scoring formulas • APIs • database • explainability • security • testing

Status: Developer-focused design proposal for the team's hackathon proof of concept. Exact implementation details have not been verified against the source repository in this document.

Important: Stack, schemas, endpoints, weights and formulas below are proposed recommendations—not claims that these features already exist or validated lending rules.

Prepared 25 October 2026

1. Executive summary

Vouch is a complementary credit-assessment layer, not a replacement for CIBIL or another authorized credit bureau score. It adds a peer-endorsement graph and a proposed Vouch Score alongside conventional credit information, repayment affordability, data quality and explanations.

The main target cases are first-time / thin-file borrowers and borrowers whose bureau score changes after a temporary financial shock. The system must not infer that a borrower is safe merely because peers vouch for them, nor infer inability to repay from absent credit history.

Assessment outputs

• **Bureau score:** display the score from an authorized source, with source and timestamp; never fabricate an official score.

• **Vouch Score (0–100):** a proposed bounded measure of outcome-aware peer endorsements, not a repayment probability.

• **Affordability:** income stability, existing obligations, estimated disposable cash flow and proposed-loan EMI.

• **Explainability:** evidence-backed positive/negative factors, missing information, confidence and scenario changes.

• **Decision support:** flag cases for review; do not automatically approve or reject loans.

2. Problem statement mapped to requirements

Pain point	System response
No / limited credit history	Treat bureau history as unavailable, not as negative behaviour. Use verified income and obligations; optionally add Vouch evidence.
Score may not reflect current circumstances	Show bureau score, affordability and data freshness as distinct signals.
Temporary financial shock	Show an unusual expense or utilization change as a reviewable event only when supported by evidence.
Black-box assessment	Return reason codes, evidence source, timestamps, uncertainty and missing data.
Endorsement gaming	Use diminishing returns, concentration limits, reciprocal-cycle checks and verified outcomes.
Fairness / incomplete data	Low data quality should produce uncertainty or review—not invented values or an automatic adverse conclusion.

3. Recommended stack

Recommended reference stack for rapid delivery. Adopt existing project choices if the current proof of concept already uses a different stack.

Layer	Recommendation	Notes
Frontend	React + TypeScript + Vite; Tailwind	Borrower onboarding, endorsement workflow and bank assessment dashboard.
API	Python 3.12 + FastAPI + Pydantic	Typed request validation, OpenAPI docs and Python scoring ecosystem.
Database	PostgreSQL	Users, consent, applications, endorsements, outcomes, assessments and audit records.
Graph MVP	PostgreSQL edge table; NetworkX for offline experiments	Use graph queries before adding a dedicated graph database.
Async jobs	Redis + RQ/Celery (optional)	Only if document parsing/report generation becomes asynchronous.
Explainability	Deterministic rules first; SHAP only for a later validated ML model	Do not let an LLM invent causes or make lending decisions.
Testing	pytest, HTTPX, React Testing Library, Playwright	Unit, API, UI, workflow and adversarial tests.

Layer	Recommendation	Notes
Deployment	Docker Compose	Repeatable local setup; production needs security and operations review.

MVP principle: a modular monolith is preferable to microservices for a hackathon. Avoid adding Kubernetes, Kafka or a graph database unless a real requirement justifies them.

4. Architecture and repository layout

```
React UI
  | HTTPS + authenticated requests
  v
FastAPI modular monolith
  ■■ authentication / authorization / consent
  ■■ borrower profile and evidence metadata
  ■■ loan application and affordability calculator
  ■■ vouch graph and score engine
  ■■ assessment + explainability report
  ■■ audit / dispute workflow
       |                |
       v                v
  PostgreSQL       private document storage (optional)
       |
  Redis / worker (optional)
```

```
backend/app/
  main.py
  api/{auth,users,applications,endorsements,assessments,reports}.py
  core/{config,security,permissions}.py
  models/       # ORM models
  schemas/      # Pydantic request/response types
  services/{affordability,vouch_score,graph_checks,explanations,assessment}.py
  repositories/
  tests/
migrations/
frontend/src/{pages,components,api,types}/
```

Keep scoring logic in pure, independently testable service functions. API handlers should validate permissions and inputs, call services, persist versioned outputs, and return structured results.

5. Data model

Entity	Important fields
User	id (UUID), display_name, verification_status, created_at.
ConsentRecord	user_id, purpose, scope, granted_at, revoked_at, policy_version.
FinancialProfile	user_id, monthly_net_income, income frequency/stability, obligations, updated_at, source and verification status.
LoanApplication	id, borrower_id, principal, annual_rate, term_months, purpose category, status.
EvidenceItem	id, owner/application, type, source, observed_at, verified_at, confidence, private storage reference, retention date.
Endorsement	id, endorser_id, recipient_id, optional application_id, statement, status, created_at, expiry.
RepaymentOutcome	application_id, period, due amount, paid amount/date, source, verification and dispute status.
Assessment	application_id, bureau score/status, Vouch Score, affordability, confidence, reason JSON, algorithm version, timestamp.
AuditEvent	actor, action, object type/id, timestamp, reason and request id; never raw sensitive documents.

Use database constraints for unique active edges and valid foreign keys. A user must not access another person's financial profile or documents by guessing an ID.

6. Vouch network

Graph model

Each user is a node. A directed edge X → Y means X endorses Y. The endorser stakes platform reputation, not money or a legal guarantee, unless a separately reviewed product explicitly defines otherwise. An endorsement should count only if the endorser is authenticated, eligible, consented and within exposure limits.

Endorsement weight

```
r_x = endorser reputation in [0,1]
q_x = verification / data-quality factor in [0,1]
e_xy = edge strength after policy checks in [0,1]
d_xy = evidence relevance factor in [0,1]
p_xy = abuse penalty in [0,1]

w_xy = clamp(r_x * q_x * e_xy * d_xy * (1 - p_xy), 0, 1)
```

Expose the factors in an internal debug view so developers can explain why an edge contributed little or was excluded. Do not infer relationships from contact lists, call logs or social media.

Proposed Vouch Score

```
W = sum of eligible incoming endorsement weights
C = smoothed weighted share of resolved outcomes that were successful
tau = saturation parameter > 0

support = 1 - exp(-W / tau)        # diminishing returns
quality = clamp(C, 0, 1)
VouchScore = round(100 * support * (0.35 + 0.65 * quality))
```

This is a design hypothesis, not a calibrated probability of repayment. Choose tau and the prior through testing. If no repayment outcome evidence exists, mark the score provisional / low-confidence rather than presenting a precise reliability claim.

Reputation updates

```
y = 1 for verified acceptable outcome; 0 for verified adverse outcome
p = expected outcome rate for a comparable case
error = y - p
delta = learning_rate * edge_weight * error * evidence_reliability
new_reputation = clamp(old_reputation + delta, 0, 1)
```

Use a dispute window, per-event caps and slow updates. A missed payment must not automatically punish every endorser; causes may be uncertain and outcomes may be disputed.

Dilution and abuse safeguards

• Diminishing returns: the 100th endorsement should contribute far less than the first few.

• One active edge per endorser-recipient pair, with expiry/cooldown policy.

• Discount reciprocal pairs and dense small clusters that repeatedly endorse each other.

• Cap total influence from a single endorser and a tightly connected cluster.

• No endorsement should guarantee a loan; strong reputation cannot override unaffordable repayments.

• Flag suspicious bursts, duplicate accounts and concentrated influence for review.

7. Affordability and loan scenarios

```
P = principal; r = monthly interest rate; n = number of monthly payments
EMI = P * r * (1+r)^n / ((1+r)^n - 1)
If r = 0: EMI = P / n

monthly_surplus = verified_net_income
                  - essential_living_cost_estimate
                  - existing_monthly_obligations
                  - proposed_EMI

debt_service_ratio = (existing_monthly_obligations + proposed_EMI)
                     / max(verified_net_income, epsilon)
```

Do not invent a precise surplus if income, obligations or living costs are missing. Return a range or 'insufficient data'. Clearly label user-declared assumptions.

• Base scenario: current income, obligations and requested loan.

• Income shock: reduce recurring income by a configurable amount.

• Emergency expense: add a one-time cost or short-term obligation and show impact.

• Loan terms: compare different principal, rate or term values.

Each scenario must show changed inputs, output differences and assumptions. It must not overwrite the base assessment.

8. How to present the combined assessment

Avoid blindly adding incomparable scores or claiming a combined score is validated. The MVP should present signals side-by-side.

```
{
  "bureau": {"status": "available | unavailable | not_authorized", "score": null},
  "vouch": {"score": 0, "status": "provisional", "confidence": "low"},
  "affordability": {"status": "sufficient | borderline | insufficient | unknown"},
  "data_quality": "high | medium | low",
  "positive_factors": [],
  "risk_factors": [],
  "missing_information": [],
  "review_flags": [],
  "algorithm_version": "assessment-v0.1"
}
```

If a demo requires a single experimental indicator, label it 'Prototype Indicator — not a lending decision'. Do not use arbitrary weights such as 60% bureau + 40% Vouch without representative repayment data, calibration, fairness tests and lender governance.

9. Explainable AI

Start with deterministic explanation templates over known fields. An LLM can rephrase structured reasons for readability, but must not infer a medical emergency from a transaction label alone or make the lending decision.

```
{
  "summary": "Affordability needs manual review",
  "positive_factors": [{
    "code": "INCOME_PRESENT",
    "text": "Verified recurring income is present",
    "source": "evidence-id",
    "as_of": "2026-10-01"
  }],
  "risk_factors": [{
    "code": "UTILIZATION_SPIKE",
    "text": "Reported revolving utilization increased",
    "source": "authorized-data-source",
    "as_of": "2026-10-10"
  }],
  "missing_information": ["Current living costs not provided"],
  "confidence": "medium",
  "limitations": ["Expense purpose not verified"],
  "algorithm_version": "assessment-v0.1"
}
```

• Separate observed facts, user statements, model outputs and hypotheses.

• Show what changed since the previous assessment and which input caused the change.

• Only call an event a medical emergency if relevant evidence supports that description; otherwise say 'large recent expense; purpose unverified'.

• Store algorithm version and a protected input snapshot or hash for auditability.

10. Suggested API routes

Route	Purpose
POST /api/v1/applications	Create application; validate amount, rate and term.
POST /api/v1/applications/{id}/evidence	Register consented evidence metadata; keep files private.
POST /api/v1/endorsements	Create endorsement; enforce identity, consent, limits and anti-abuse rules.
GET /api/v1/users/{id}/vouch-summary	Return score, confidence and eligible count; hide private endorsement details.

Route	Purpose
POST /api/v1/applications/{id}/assessments	Run affordability and Vouch calculations; persist versioned result.
GET /api/v1/assessments/{id}	Return scores, affordability, explanations, missing fields and review flags.
POST /api/v1/assessments/{id}/scenarios	Recalculate with changed assumptions without overwriting base result.
POST /api/v1/endorsements/{id}/dispute	Open review for a disputed endorsement/outcome.
GET /health	Health check without exposing configuration or secrets.

Every endpoint needs authentication and object-level authorization. Apply rate limits and request validation. Use consistent error responses and request IDs.

11. Security and privacy

• Use HTTPS, Argon2id/bcrypt password hashing, safe token handling, RBAC and rate limits.

• Collect only data needed for a stated purpose; record explicit consent and support revocation according to policy.

• Encrypt sensitive data in transit and at rest; store secrets outside source control.

• Never log passwords, tokens, raw bank statements or full identity numbers.

• Define retention/deletion, correction and dispute workflows. Keep document storage private.

• Do not scrape contacts, call logs or social media to discover endorsers; endorsements are explicit platform actions.

• Use synthetic demo data. Production deployment needs legal, compliance, security and model-risk review.

India regulatory note: RBI's Digital Lending Directions, 2025 include requirements concerning borrower creditworthiness, disclosures, need-based data collection, explicit consent and data handling for covered regulated entities and their lending-service arrangements. Applicability depends on the actual business model; obtain qualified compliance review before production. Official source: https://www.rbi.org.in/Scripts/BS_ViewMasDirections.aspx?id=13156

12. Testing and evaluation

Unit tests

• EMI formula: zero and positive rates, edge cases and invalid inputs.

• Vouch score: no edges, one edge, many edges, low-quality endorser, expired edge and no outcomes.

• Graph rules: block self-vouch and duplicate edges; discount reciprocal/circular groups.

• Disputed/unverified outcomes must not cause final reputation penalties.

• Missing income/obligation data returns unknown/review, not fabricated zero values.

Adversarial / integration tests

• A coordinated ring cannot raise scores without limit.

• A high-degree endorser cannot dominate the assessment.

• Unauthorized users cannot access other profiles, evidence, endorsements or reports.

• Consent revocation blocks future use as required by policy.

Metrics

• If a probability model is later trained: calibration, Brier score/log loss, stability and error rates.

• Network robustness: score inflation under simulated collusion, concentration of influence, node-removal sensitivity.

• Explainability: traceability to inputs, factual consistency and user comprehension.

• Fairness: evaluate error and review rates across appropriately sampled groups where lawful; investigate proxies and small-sample uncertainty.

A hackathon demo with synthetic data can demonstrate workflow and abuse-resistance tests; it cannot establish real-world lending performance.

13. Implementation roadmap

Phase	Deliverable	Acceptance test
1. Foundation	Auth, roles, profile, consent and database migrations	New teammate can run app; permissions tested.
2. Affordability	Loan form, income/obligation fields, EMI calculator	Known EMI examples pass; missing data is explicit.
3. Vouch graph	Create/revoke endorsements, score v0.1, anti-abuse checks	Self/duplicate edges blocked; diminishing returns tested.
4. Assessment	Side-by-side bureau status, Vouch, affordability and reasons	Each displayed factor has a source or explicit assumption.
5. Scenarios	Change income/loan terms and compare results	Base result remains unchanged; diff shown.
6. Demo hardening	Audit log, synthetic seed data, tests, Docker README	Reproducible demo with no real PII or secrets.

14. Developer definition of done

• Bureau score, Vouch Score and affordability are separate signals.

• No credit history is not treated as negative repayment behaviour.

• Every endorsement has actor, timestamp, status and abuse checks.

• Score is bounded, uses diminishing returns and stores algorithm version.

• Verified outcomes update reputation through a documented, auditable rule.

• Assessment explains evidence, missing information, uncertainty and scenario changes.

• Consent and access controls protect personal and financial data.

• README documents setup, environment variables, migrations, tests, seed data and known limitations.

• All features are marked implemented / partial / planned after code inspection.

15. References and implementation caveat

RBI — Digital Lending Directions, 2025: https://www.rbi.org.in/Scripts/BS_ViewMasDirections.aspx?id=13156

RBI website / publications: https://website.rbi.org.in/

PageRank / graph ranking overview: https://doi.org/10.1145/3760438 . PageRank is inspiration for weighting incoming links; it is not a validated credit-risk formula.

This document is an implementation blueprint. Before using it as an accurate description of the built prototype, inspect the team's repository and replace recommendations with actual package names, routes, schemas, UI labels and a truthful implemented/partial/planned status matrix.

VOUCH

Developer Master Documentation — Version 2

Expanded scenario analysis: where conventional credit scores can be insufficient

Purpose of this revision

This revision preserves the previous developer master document and appends a detailed scenario chapter explaining the four core cases and the cross-cutting network and explainability cases described by the team. It is written to align developers, designers, and pitch presenters on the problem the prototype is intended to address.

> **Status:** Design proposal for a hackathon proof of concept. The existence of a proof of concept was stated by the team; the detailed algorithms and workflows below remain proposals unless confirmed in the source code.

Core positioning

Vouch is a complementary assessment layer intended to improve the usefulness of conventional credit information in cases where that information is missing, incomplete, stale, or insufficient to explain a borrower’s present circumstances. It does not replace CIBIL, recreate an official bureau score, guarantee repayment, or make a loan decision by itself.

17. Core scenarios: problem, proposed response, and developer implications

These scenarios are central to the product definition—not optional edge cases. The system should preserve the distinction between bureau history, present repayment capacity, peer-backed reputation, and uncertainty.

Scenario 1 — First-time borrower / no bureau score

A young adult has recently started a job and applies for a first loan. There may be no meaningful credit history from which a conventional bureau score can be derived. Missing history is not evidence of irresponsible repayment, but it is also not proof that the applicant can repay.

Signals to collect (with consent): Employment or offer confirmation; recent payslips or other verified income evidence; income frequency and stability; existing loans and recurring obligations; essential expenses where available; requested principal, rate, and term; eligible peer endorsements.

Proposed system behaviour: Represent the bureau status explicitly as unavailable/no-file rather than assigning a fabricated low score. Calculate affordability only from known inputs. Display the Vouch Score separately, along with the number and quality of eligible endorsements, evidence coverage, and confidence. If critical data is missing, return 'insufficient information' and specify what would resolve it.

Example explanation: “No usable bureau score was available. Verified income evidence is present, but current living costs are incomplete. Eligible peer endorsements provide additional relational evidence; they do not establish affordability. Review the missing expense and obligation information before making a decision.”

Developer requirements: Support bureau_status values such as AVAILABLE, NO_FILE, UNAVAILABLE, and NOT_AUTHORIZED. Do not convert null bureau values to zero. Validate income and loan inputs; store provenance, verification state, as-of date, and consent. Include tests for no bureau score, no vouches, no income proof, and conflicting income evidence.

What this scenario demonstrates: Vouch can add a structured source of evidence when traditional repayment history is absent, while keeping the affordability question separate.

Scenario 2 — Temporary financial shock / sudden bureau-score decline

A borrower has a stable job and previously managed obligations, then faces a large unexpected expense—for example, a medical expense that the borrower reports was not covered by insurance. The borrower uses a large share of available revolving credit. The bureau score may decline, but the score change alone cannot determine whether current repayment capacity has materially changed.

Signals to compare: Prior and current income; existing and newly added obligations; revolving utilization or balance changes where lawfully available; payment history; available cash-flow evidence; proposed EMI; reported reason for the change; evidence supporting that reason; dates of all relevant observations.

Proposed system behaviour: Create a before/after comparison and flag a material change for review. Distinguish observed facts from the borrower's explanation. Do not automatically preserve the borrower's risk status because income appears stable, and do not automatically classify the borrower as unable to repay because utilization rose. Recompute affordability using current obligations and the proposed loan terms.

Explainability rule: Only label the event a medical emergency if suitable evidence supports that description and collection/use is lawful and consented. Otherwise use neutral wording such as “large recent expense; purpose reported but not verified.” Avoid inferring sensitive health information from transaction text.

Developer requirements: Version assessments; keep an immutable base assessment and create a new assessment for changed inputs. Store reason codes such as UTILIZATION_SPIKE, OBLIGATION_INCREASE, INCOME_STABLE, and PURPOSE_UNVERIFIED. Support scenario simulation for one-time expense, new debt, and income changes. Route conflicting or weak evidence to human review.

What this scenario demonstrates: A score change is a signal to investigate, not a complete explanation of the borrower's current ability to pay.

Scenario 3 — Peer-to-peer vouching and reciprocal incentives

A person with an established profile and both an eligible bureau record and Vouch reputation endorses another applicant. The endorser stakes platform reputation on the endorsement. The recipient gains additional relational evidence; the endorser may later gain or lose reputation based on verified outcomes, subject to the defined policy and a dispute process.

Eligibility: Any authenticated platform user who meets the system's eligibility policy and has an existing usable bureau profile and Vouch Score may be allowed to vouch. A strong score should not automatically make a user eligible for unlimited endorsements.

Proposed system behaviour: Record the directed edge endorser → recipient, timestamp, consent, status, optional application scope, commitment/exposure limit, and expiry. Calculate an endorsement's contribution using endorser reliability, evidence quality, relationship/edge eligibility, and commitment utilization. Show lenders aggregated evidence, not private social details.

Incentive mechanism: A verified successful repayment outcome may contribute positively to the endorser's reputation; a verified adverse outcome may contribute negatively only under a clearly defined, proportionate policy. Disputed, unverified, or causally ambiguous outcomes must not trigger automatic penalties. Vouching is a reputation stake, not a financial guarantee unless a separate legally reviewed contract explicitly says otherwise.

Developer requirements: Prevent duplicate active edges; rate-limit endorsement creation; track total active commitments; support expiry, withdrawal rules, disputes, audit events, and idempotent outcome processing. Keep endorsement creation separate from outcome verification and score recalculation.

What this scenario demonstrates: Vouching creates a reputation-based incentive to make endorsements thoughtfully, rather than merely collecting endorsements.

Scenario 4 — Dilution, influential endorsers, and endorsement-ring attacks

A highly influential or wealthy person receives a very large number of endorsements, or a small cluster of users repeatedly endorses one another. Without safeguards, raw endorsement counts could be inflated and the system could become a popularity contest.

Threats: Endorsement farming; reciprocal rings; duplicate or synthetic accounts; many endorsements controlled by one actor; correlated endorsers; rapid endorsement bursts; users taking on more commitments than their reputation budget supports.

Proposed system behaviour: Use diminishing returns so each additional endorsement has less marginal influence. Cap an individual endorser's contribution and total exposure. Detect reciprocal patterns and dense clusters for review. Require verified identities appropriate to the use case and eligible, explicit endorsements. Do not use wealth, fame, follower count, or raw endorsement volume as a proxy for repayment capacity.

Illustrative graph weight: For an eligible edge $x \to y$, define $w_{xy} = \text{clamp}(R_x \times Q_x \times E_{xy} \times D_{xy} \times (1 - P_{xy}), 0, 1)$, where R is endorser reputation, Q is evidence/verification quality, E is edge eligibility/strength, D is relevance, and P is an abuse penalty. Each factor is an experimental design input that needs calibration and auditability.

Illustrative diminishing-return score: Let W be the sum of eligible effective edge weights and $\tau > 0$ a calibration parameter. A simple support term is $S = 100 \times (1 - \exp(-W/\tau))$. This limits score growth as endorsements accumulate. It is not a probability of repayment and must not be treated as a validated credit score.

Developer requirements: Log exclusion reasons; expose safe internal diagnostics; create synthetic attack graphs for tests; test score movement after duplicates, reciprocal edges, and one dominant endorser are added. Tune thresholds using representative data and document false-positive risks.

What this scenario demonstrates: A credible network score must measure qualified evidence, not popularity, and must remain resistant to coordinated manipulation.

18. Cross-cutting case — Explainability and evidence quality

Explainability is not a decorative AI summary added after scoring. It is a structured record of what the system observed, what was reported, what it inferred, how each factor affected the assessment, and what remains uncertain.

Required explanation fields

• Summary and assessment timestamp.
• Bureau status, source, and as-of date (where available and authorized).
• Vouch Score, status, confidence, eligible endorsement count, and score version.
• Affordability result, assumptions, and missing inputs.
• Positive factors and risk factors, each with a reason code, source, date, and verification status.
• Reported circumstances separated from verified facts.
• Uncertainty, contradictory evidence, limitations, and next steps for manual review.

Example output

> “The bureau score has declined and revolving utilization has increased. Available records indicate recurring income may be stable, but the purpose of the recent expense is not verified and current living costs are incomplete. The applicant has eligible peer endorsements. The evidence does not establish that the expense was a medical emergency or prove that the requested loan is affordable. Review current obligations and residual cash flow.”

AI implementation boundary

For the MVP, calculate scores with deterministic, versioned code and generate reason codes from known inputs. An LLM may turn those structured reasons into clearer language, but must not invent facts, classify sensitive events without evidence, choose numerical score inputs, or make the lending decision. Keep source provenance and uncertainty visible.

19. Scenario-to-component map

Scenario	Core modules	Primary tests
No credit history	Evidence intake; bureau status; affordability; Vouch graph; uncertainty	Null bureau score; no vouches; missing income; conflicting evidence
Temporary financial shock	Change detection; before/after assessment; scenario engine; explanation provenance	Stable income + new debt; income loss; unverified event purpose
Peer vouching	Identity/consent; endorsement API; commitments; outcome ingestion; reputation updates	Duplicate edge; disputed outcome; expiry; repeated event; unauthorized access
Dilution / manipulation	Graph analytics; contribution caps; diminishing returns; abuse flags	Endorsement ring; dominant endorser; duplicate accounts; burst activity
Explainability	Reason-code engine; evidence provenance; versioned assessments; review queue	Unsupported emergency claim; stale evidence; missing inputs; reproducibility

20. Pitch-stage framing

Problem: Conventional credit information can be incomplete or may not fully explain a borrower's current circumstances. A missing score is not the same as a negative repayment record, and a score decline does not by itself explain affordability.

Solution: Keep the bureau score, peer-backed Vouch Score, affordability analysis, and evidence-based explanations as distinct signals. Use endorsements to add relational evidence, not to override affordability.

Trust safeguards: Reputation staking, verified outcomes, diminishing returns, contribution limits, collusion checks, explicit consent, and transparent uncertainty.

Honest prototype claim: The team reports having a proof of concept. Before presenting any particular API, model, formula, integration, or automated feature as implemented, verify it against the repository. Treat formulas in this document as experimental proposals until tested and calibrated.

21. Acceptance criteria for the scenario module

1. The system never treats missing bureau history as a numeric zero or automatically as negative behaviour.
2. A bureau-score change triggers a review of current evidence rather than an unsupported causal claim.
3. The lender view keeps bureau score, Vouch Score, affordability, data quality, and confidence distinguishable.
4. Vouch score cannot override a clear affordability shortfall or lender policy.
5. Every endorsement has eligibility, consent, timestamp, status, and auditable contribution logic.
6. Unverified or disputed repayment outcomes do not automatically change reputation.
7. Adding many low-quality or correlated endorsements has bounded marginal impact.
8. Every explanation identifies evidence, missing data, limitations, and the assessment version.
9. Base assessments are preserved when hypothetical scenarios are run.
10. Demo and pitch language clearly distinguishes implemented features from proposed design.

End of scenario expansion. This chapter is appended to the prior developer master documentation; it does not replace the original technical stack, data model, API proposal, formulas, security guidance, or roadmap.